NB IMCQ 2
Neuroscience & Anatomy

1. A 58-year-old man comes to the physician because of a 2-year history of progressively worsening problems with movement. Neurologic examination shows resting tremor, rigidity, bradykinesia, and postural instability. The neurotransmitter most directly deficient in this patient belongs to which of the following classes?

A. Catecholamine
B. Indoleamine
C. Excitatory amino acid
D. Inhibitory amino acid
E. Neuropeptide

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SOM.MK.I.BPM2.1.NB.1.NSCI.0113 Describe the nuclei and projections and receptors tied to catecholamines and serotonin

2. A histopathological sample of brain tissue is subjected to postmortem laboratory examination. The deceased patient had a 5-year history of severe dementia that was treated with acetylcholinesterase inhibitors. Which of the following nuclei is most likely degenerated in the sample extracted from this patient?

A. Ventral tegmental area
B. Raphe Magnus nucleus
C. Locus coeruleus
D. Substantia nigra
E. Basal nucleus of Meynert

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SOM.MK.I.BPM2.1.NB.1.NSCI.0111 Describe the nuclei and projections and receptors tied to acetylcholine

3. A 46-year-old woman is brought to the emergency department one hour after she was involved in a motor vehicle collision. A CT scan of the head shows a piece of glass has severed one of the nerves located in the orbit. After successful removal of the glass, physical examination shows the corneal reflex is absent. Which of the following nerves most likely carries the afferent fibers?

A. Oculomotor
B. Great auricular
C. Facial
D. Nasociliary
E. Infraorbital
F. Inferior alveolar

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Describe the nerves that carry general sensation from the eyeball.

SOM.MKII.BPM2.1.NB.1.ANAT.0024

4. A 55-year-old women comes to the physician because of a 2-cm, firm, non-tender nodule in the upper portion of the right lobe of her thyroid gland. She also has mild dysphagia. Her thyroid function tests are normal and fine needle aspiration of the nodule shows a benign follicular lesion. During surgery to remove the nodule, the resident accidentally lesions a nerve lying next to the artery he was trying to ligate. Damage to this nerve typically results in a monotone voice. The artery that was being ligated is most likely a direct branch of which of the following arteries?

A. Superior laryngeal
B. External carotid
C. Internal carotid
D. Thyrocervical trunk
E. Subclavian

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SOM.MKII.BPM2.1.NB.1.ANAT.0074
Describe the branches of the external carotid artery and their areas of distribution

5. A 25-year-old man is brought to the physician because of a 2-week history of being unable to rotate his head to the left side. He sustained a stab wound to his neck just posterior to the sternocleidomastoid during a domestic dispute 3 weeks ago. Which of the following structures is most likely to accompany the injured structure as it exits the cranial cavity?

A. Vertebral artery
B. Basilar artery
C. Transverse cervical artery
D. Great auricular nerve
E. Lesser occipital nerve
F. Internal jugular vein
G. External jugular vein

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Describe the extracranial course of the accessory nerve and what it supplies

SOM.MKII.BPM2.1.NB.1.ANAT.0194

6. A 15-year-old boy is brought to the physician by his mother because of a 5-day history of severe headaches and a feeling of heaviness in the middle part of his face. He has a history of a recent bacterial upper respiratory tract infection. Imaging studies show fluid accumulation in the paranasal sinuses. Which of the following statements best describes the normal drainage of the paranasal sinuses?

A. The frontal sinus drains into the superior meatus
B. The maxillary sinus drains into the middle meatus
C. The anterior ethmoid air cells drain into the superior meatus
D. The posterior ethmoid air cells drain into the sphenoethmoidal recess
E. The sphenoid sinus drains into the inferior meatus

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Describe the paranasal sinuses, the bones in which they are found and where each opens into the nasal cavity.
SOM.MKII.BPM2.1.NB.1.ANAT.0101

7. A 40-year-old man comes to the physician one hour after he sustained blunt force trauma to the post-auricular region of the head. Imaging studies show multiple fractures. One lesion affects fibers arising from the geniculate ganglion that course across the anterior surface of the petrous bone toward the roof of the foramen lacerum. Which of the following will most likely be observed on physical examination?

A. Dry eyes and nasal mucous glands
B. Drooling from the side of the mouth and hyperacusis
C. Depressed nasolabial folds and decreased salivation
D. Dry paranasal sinuses and loss of taste on the anterior two-thirds of the tongue
E. Compromised balance and equilibrium

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Describe the geniculate ganglion, including its composition, location and branches
SOM.MKII.BPM2.1.NB.1.ANAT.0171

8. A 17-year-old girl is brought to the physician by her mother because of a 3- week history of continuous nose bleeds. Physical examination shows a hyperemic area on the left anterior nasal septum, an area of anastomosis of blood vessels where the bleeding usually originates. Which of the following arteries are most likely anastomotic vessels of the affected area?

A. Sphenopalatine and anterior labial
B. Lesser palatine and superior labial
C. Anterior ethmoidal and superior labial
D. Superior and anterior labial
E. Sphenopalatine and middle meningeal

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Explain the clinical significance of Kiesselbach's area in relation to epistaxis

9. A 40-year-old woman comes to the physician because of 3-month history of reduced salivation in the mouth. Physical examination shows notable facial asymmetry in the pre-auricular region on the left side. Imaging study of the head shows a cystic mass on the medulla just inferior to the pontomedullary junction. Which of the following nuclei is most likely involved?

A. Nucleus of Trigeminal nerve
B. Dorsal motor nucleus
C. Solitary nucleus
D. Inferior salivatory nucleus
E. Superior salivatory nucleus

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Describe the locations of the nuclei of the facial nerve and their associated structures
SOM.MKII.BPM2.1.NB.1.ANAT.0167

10. A 45-year-old woman undergoes thyroidectomy for a large multinodular goiter. Postoperatively, she develops hoarseness and difficulty swallowing. Physical examination shows a weak cough and impaired elevation of the pharynx during swallowing. Which of the following muscles is most likely affected?

A. Cricopharyngeus
B. Inferior pharyngeal constrictor
C. Palatopharyngeus
D. Stylopharyngeus
E. Superior pharyngeal constrictor

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SOM.MKII.BPM2.1.NB.1.ANAT.0125
List the general attachments, innervations and functions of the muscles of the pharynx

11. A 13-year-old boy is brought to the physician by his mother because of a 2-day history of severe headaches and a feeling of heaviness in the mid-part of his face. He has a history of a recent bacterial upper respiratory tract infection. Imaging studies show fluid accumulation and inflammation in the maxillary sinuses. Which of the following nerves is most likely to mediate the pain from the affected sinus?

A. Zygomaticofacial
B. Anterior ethmoidal
C. Supratrochlear
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SOM.MK.I.BMP2.1.NB.1.ANAT.0103
Describe the sensory and parasympathetic innervations of the paranasal sinuses.

12. A 15-year-old boy is brought to the physician by his parents because of a 2-week history of being unable to rotate his head to the left side. He sustained a stab wound to his neck just posterior to the sternocleidomastoid during a fight at school 3 weeks ago. Which of the following structures most likely accompanies the injured structure as it enters the cranial cavity?

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SOM.MK.I.BMP2.1.NB.1.ANAT.0067 Describe the general attachments, innervations and actions of the muscles that are found in the floor of the posterior triangle

13. A 13-year-old boy is brought to the emergency department by his mother because of a 3-day history of sharp pain in his left ear. Otoscopic examination shows bulging of the tympanic membrane. Diagnosis of middle ear infection is made. Which of the following is most likely the cell body location of the sensory fibers from the affected region?

A. Trigeminal ganglion
B. Geniculate ganglion
C. Submandibular ganglion
D. Pterygopalatine ganglion
E. Superior (jugular) ganglion
F. Inferior (petrous) ganglion

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SOM.MKII.BPM2.1.NB.1.ANAT.0182

Give the name and location of the ganglia formed by the glossopharyngeal nerve

14. A 65-year-old man comes to the physician because of a 2-week history of numbness of the lower lip and left side of his face anterior to his ear and difficulty chewing his food. The nerve responsible for innervation to the affected region most likely passes through which of the following foramen?

A. Foramen ovale
B. Foramen rotundum
C. Internal acoustic meatus
D. Stylomastoid foramen
E. Mandibular foramen
F. Foramen spinosum
G. Inferior orbital fissure

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Describe the foramina through which each of the divisions of the trigeminal nerve pass through.
SOM.MKII.BPM2.1.NB.1.ANAT.0163

15. A 10-year-old boy is brought to the physician by his parents because of a 2-day history of fever and intense pain around his ear. Physical examination shows a large swelling antero-inferior to the ear that is tender and warm to the touch. He is unable to tightly close his left eye and cannot frown his forehead on the same side. Which of the following nerves is most likely affected due to this injury?

A. Supratrochlear
B. External nasal
C. Zygomaticotemporal
D. Temporal
E. Marginal mandibular
F. Cervical

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SOM.MKII.BPM2.1.NB.1.ANAT.0082 List the locations, innervations and actions of the major muscle groups of facial expression

16. A 32-year-old man comes to the physician because of injuries to the face following a motor vehicle collision. Physical examination shows swelling and bruising on the right half of his face, including the eye. Sensory testing shows decreased sensation in the right midface area including the lower eyelid and the upper lip. There is no sensory loss associated with the nasal or palate mucosa lacrimation and secretions from the palate and nasal mucosa are also normal. Motor testing shows no weakness of the facial muscles. Which of the following describes the most likely injury resulting in these findings?

A. Fracture of the medial wall of the orbit
B. Fracture affecting the foramen rotundum
C. Fracture of the pterygoid plate affecting the pterygopalatine ganglion
D. Fracture of the inferior wall of the orbit
E. Fracture affecting the superior orbital fissure

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C. Fracture of the pterygoid plate affecting the pterygopalatine ganglion
D. Fracture of the inferior wall of the orbit
E. Fracture affecting the superior orbital fissure

Describe fractures of the orbit and their clinical consequences including extraocular muscle impingement
SOM.MKII.BPM2.1.NB.1.ANAT.0240